

Top 100 Farming & Agriculture Questions with Detailed Answers

A. Basic Farming & Farm Planning

1. What is farming?

Farming is the practice of producing crops, livestock, or other agricultural products using land, water, labour, inputs, and management. It includes activities such as land preparation, sowing, irrigation, nutrient management, pest control, harvesting, storage and marketing.

2. What are the major types of farming?

Major types include subsistence farming, commercial farming, intensive farming, extensive farming, organic farming, natural farming, mixed farming, plantation farming, rainfed farming, irrigated farming, and integrated farming.

3. How should a farmer choose a crop?

Crop selection should consider:

- Soil type and fertility
- Climate and rainfall
- Irrigation availability
- Growing season
- Availability of quality seed
- Pest and disease risks
- Labour and input costs
- Expected market demand and price
- Farmer's available land and resources

A crop that grows well but has poor market access may not be financially attractive.

4. What factors affect crop production?

Important factors include seed quality, soil fertility, water availability, temperature, sunlight, rainfall, nutrient management, weeds, pests, diseases, planting time, crop variety, farm management and harvesting practices.

5. What is a cropping system?

A cropping system describes how crops are arranged and grown on a farm over time. Examples include monocropping, multiple cropping, intercropping, sequential cropping and crop rotation.

6. What is crop rotation?

Crop rotation means growing different crops on the same land in a planned sequence across seasons or years. For example, a cereal followed by a pulse can diversify nutrient use and help break some pest and disease cycles.

ICAR guidance specifically recommends changing rice-based sequences to rice-pulse or rice-oilseed systems in appropriate situations. ([ICAR-CIWA](#))

7. What is intercropping?

Intercropping means growing two or more crops together in the same field during part or all of their growing periods. Proper combinations can improve resource use and provide diversification.

8. What is mixed farming?

Mixed farming combines crop production with livestock or other agricultural enterprises. Crop residues can potentially support livestock, while properly managed livestock manure can contribute nutrients and organic matter to fields.

9. What is sustainable agriculture?

Sustainable agriculture aims to maintain productive farming while conserving soil, water, biodiversity and other resources for the future. It emphasizes efficient input use, soil health, responsible water management and long-term farm resilience.

10. What is climate-smart agriculture?

Climate-smart agriculture is an approach that aims to address food production, climate adaptation and, where possible, mitigation together. The exact practices depend on local conditions rather than one universal recipe. ([FAOHome](#))

B. Soil & Soil Health

11. Why is soil important for farming?

Soil provides plants with physical support, water, air and nutrients. Healthy soil also contains biological organisms and organic matter that influence nutrient cycling, structure and water retention.

12. What are the major types of agricultural soil?

Common soil classifications include sandy, clayey, loamy, alluvial, black, red and lateritic soils. Their properties differ in drainage, water-holding capacity, nutrient availability and suitability for different crops.

13. Which soil is best for agriculture?

There is no single "best" soil for every crop. Loamy soils are often favourable for many crops because they can provide a balance between drainage, aeration and water retention, but crop-specific suitability depends on soil properties and climate.

14. What is soil testing?

Soil testing is the analysis of a soil sample to determine characteristics such as pH, electrical conductivity and available nutrients. It helps farmers make more informed fertilizer and soil-management decisions.

15. Why should farmers test their soil?

Testing can identify nutrient deficiencies or excesses and help avoid unnecessary fertilizer application. ICAR has emphasized soil-test-based fertilizer application as a way to improve nutrient-use efficiency and maintain long-term soil fertility. ([Indian Council of Agricultural Research](#))

16. What is soil pH?

Soil pH indicates how acidic or alkaline the soil is. Different crops have different preferred pH ranges, and extreme pH can affect nutrient availability.

17. What is soil organic matter?

Soil organic matter consists of decomposed plant, animal and microbial material. It contributes to soil structure, nutrient cycling and water-holding capacity.

18. How can soil fertility be improved?

Possible approaches include:

- Applying nutrients based on soil tests
- Adding suitable organic matter
- Growing legumes
- Crop rotation
- Returning suitable crop residues
- Reducing erosion
- Avoiding unnecessary nutrient application

19. What is soil erosion?

Soil erosion is the removal of soil by water, wind or other forces. It can remove fertile topsoil and reduce long-term productivity.

20. How can farmers control soil erosion?

Methods include maintaining plant cover, contour farming where appropriate, mulching, terracing on suitable slopes, maintaining vegetation barriers, reducing excessive soil disturbance and managing runoff.

C. Seeds & Sowing

21. How should farmers select good seeds?

Farmers should consider:

- Crop and variety suitability
- Certified or quality-assured seed where available
- Germination quality
- Purity

- Disease-free condition
- Local adaptation
- Appropriate maturity duration

22. Why is seed treatment important?

Seed treatment can protect seeds and young seedlings against certain seed-borne or soil-borne pests and diseases. The treatment should be appropriate for the crop and specific problem.

23. What is seed germination?

Germination is the process in which a viable seed begins developing into a seedling under suitable conditions of moisture, temperature, oxygen and other requirements.

24. What is direct sowing?

Direct sowing means placing seeds directly in the field where the crop will grow. Crops such as many cereals and pulses can be established this way depending on the production system.

25. What is transplanting?

Transplanting involves raising seedlings in a nursery and then moving them to the main field. Rice, tomato, chilli and some vegetable crops are commonly transplanted.

26. What is the best time to sow a crop?

The appropriate sowing time depends on crop, variety, region, temperature, rainfall, soil moisture and cropping system. Sowing calendars should therefore be location-specific.

27. What happens if sowing is delayed?

Delayed sowing can expose a crop to unsuitable temperatures or rainfall later in its life cycle and may reduce yield. The effect varies considerably by crop and region.

28. What is seed rate?

Seed rate is the amount of seed used to establish a crop over a specified area, such as kilograms per hectare. It depends on crop, variety, seed size, germination rate, planting method and target plant population.

29. What is plant spacing?

Plant spacing is the distance maintained between plants and/or rows. Appropriate spacing allows crops to access light, water and nutrients while influencing disease and weed pressure.

30. What is a nursery in agriculture?

A nursery is an area where seedlings are raised and managed before being transplanted into their final growing location.

D. Irrigation & Water Management

31. What is irrigation?

Irrigation is the artificial application of water to agricultural land when natural rainfall does not provide sufficient moisture.

32. What are the major irrigation methods?

Common methods include:

- Flood irrigation
- Furrow irrigation
- Basin irrigation
- Sprinkler irrigation
- Drip irrigation

The appropriate method depends on crop, soil, terrain, water availability and cost.

33. What is drip irrigation?

Drip irrigation delivers water close to the plant root zone through emitters. It can improve water-use efficiency when properly designed, operated and maintained.

34. What is sprinkler irrigation?

Sprinkler irrigation distributes water through pressurized nozzles, producing droplets similar to rainfall. It can be useful where field conditions and crop requirements make overhead irrigation appropriate.

35. How can farmers save irrigation water?

Useful approaches include:

- Drip or sprinkler systems where appropriate
- Irrigating according to crop demand
- Mulching
- Maintaining irrigation infrastructure
- Rainwater harvesting
- Improving soil organic matter
- Avoiding unnecessary irrigation

FAO identifies rainwater harvesting, soil-moisture conservation and improved on-farm water distribution as important approaches for improving water productivity in rainfed systems.

([FAOHome](#))

36. How often should a crop be irrigated?

There is no universal interval. Irrigation frequency depends on crop growth stage, soil type, weather, rainfall, root depth and irrigation method.

37. What is waterlogging?

Waterlogging occurs when excessive water fills soil pores and reduces the oxygen available to roots. It can severely affect crops that are not adapted to saturated conditions.

38. What is rainfed farming?

Rainfed farming depends primarily on rainfall rather than regular irrigation. Management focuses heavily on conserving available soil moisture and selecting suitable crops and varieties.

39. How can farmers manage drought?

Strategies may include drought-tolerant or early-maturing varieties, moisture conservation, appropriate sowing times, rainwater harvesting, mulching and efficient irrigation where available.

40. How can rainwater be harvested on farms?

Depending on local conditions, farmers can use farm ponds, contour structures, check dams, storage tanks, field bunds and other appropriate water-harvesting systems.



E. Manures & Fertilizers

41. What is fertilizer?

Fertilizer is a material applied to supply plant nutrients. Fertilizers may provide nutrients such as nitrogen, phosphorus, potassium or micronutrients.

42. What is manure?

Manure is an organic material derived from sources such as animal waste and decomposed plant material. It can add organic matter and nutrients to soil.

43. What is compost?

Compost is decomposed organic material produced through controlled biological decomposition of suitable organic wastes.

44. What is the difference between manure and fertilizer?

Manure generally contains lower and more variable nutrient concentrations while adding organic matter. Fertilizers are usually more concentrated and provide specified nutrients. Both may have roles in integrated nutrient management.

45. What are NPK fertilizers?

NPK refers to nitrogen (N), phosphorus (P) and potassium (K), three major plant nutrients.

- Nitrogen supports vegetative growth.
- Phosphorus plays important roles in energy transfer and root/reproductive development.
- Potassium supports numerous physiological processes and plant stress regulation.

46. Why is nitrogen important for crops?

Nitrogen is essential for plant growth and is a component of proteins, nucleic acids and chlorophyll. Both nitrogen deficiency and excessive nitrogen application can cause problems.

47. Why is phosphorus important?

Phosphorus participates in energy transfer and is important for root development and reproductive processes.

48. Why is potassium important?

Potassium regulates several physiological processes, including water relations and enzyme activity, and contributes to plant resilience.

49. What are micronutrients?

Micronutrients are nutrients required by plants in relatively small quantities. Examples include zinc, iron, boron, manganese, copper, molybdenum and chlorine.

50. Can excessive fertilizer harm crops?

Yes. Excessive or inappropriate fertilizer application can cause nutrient imbalance, salt-related problems, economic losses and environmental pollution. Fertilizer decisions should ideally be based on soil tests, crop requirements and local recommendations. ([Indian Council of Agricultural Research](#))

F. Weeds

51. What are weeds?

Weeds are unwanted plants growing where they interfere with the intended crop.

52. Why are weeds harmful?

Weeds can compete with crops for water, nutrients, light and space. Some can also harbour pests and diseases or interfere with harvesting.

53. How can weeds be controlled?

Methods include:

- Hand weeding
- Mechanical cultivation
- Mulching
- Crop rotation
- Appropriate herbicides
- Cover crops
- Good crop establishment

Integrated weed management combines suitable approaches rather than relying on one method.

54. What is mulching?

Mulching means covering soil with materials such as crop residues, leaves or suitable manufactured materials. It can help conserve moisture and suppress some weeds.

55. When should weeds be controlled?

Weed control is particularly important during the crop's critical competition period. The exact period varies by crop and production system.

G. Pests & Diseases

56. What is a crop pest?

A crop pest is an organism that damages crops or reduces their quality or productivity. Insects are common examples, but pests can include mites, rodents and other organisms.

57. What is a plant disease?

A plant disease is a harmful abnormal condition caused by infectious agents such as fungi, bacteria, viruses or other pathogens, or sometimes by non-infectious environmental factors.

58. Why are crop leaves turning yellow?

Yellowing can result from many causes, including nutrient deficiency, excessive water, root problems, disease, pests, unsuitable pH or natural ageing. A diagnosis requires information about the crop, symptoms, field conditions and growth stage.

59. Why are leaves curling?

Leaf curling can result from insect feeding, viral diseases, water stress, nutrient problems, herbicide injury, temperature stress or other causes. The exact cause should be identified before treatment.

60. How can farmers identify crop pests?

Farmers can inspect leaves, stems, roots, flowers and fruits for insects, eggs, larvae, feeding damage, webbing, holes, discoloration or other symptoms. Good diagnosis should consider the entire field rather than one damaged plant.

61. What is integrated pest management?

Integrated Pest Management (IPM) combines monitoring, prevention, cultural practices, biological control and other appropriate methods to manage pests while reducing unnecessary pesticide use.

62. What is biological pest control?

Biological control uses natural enemies or beneficial organisms to suppress pests. Examples include predators, parasitoids and certain microorganisms.

63. What is a beneficial insect?

A beneficial insect contributes to agriculture by pollinating plants or naturally suppressing pests. Bees and many predatory insects are examples.

64. Why should farmers monitor pests before spraying?

Monitoring helps determine whether a pest is actually present and whether intervention is justified. It can reduce unnecessary pesticide applications and associated costs and risks.

65. What precautions should be taken when using pesticides?

Farmers should follow the product label and local agricultural recommendations, use only legally approved products for the intended crop and pest, observe required protective measures and respect waiting periods and application instructions.

H. Major Crop Production

66. How is rice cultivated?

Rice production generally involves variety selection, nursery/direct establishment, field preparation, planting, nutrient and water management, weed and pest management, and harvesting. Practices vary greatly between irrigated, rainfed, transplanted and direct-seeded systems.

67. How is wheat cultivated?

Wheat cultivation generally includes selecting a suitable variety, preparing the field, timely sowing, appropriate seed rate and spacing, nutrient management, irrigation according to crop need, weed and pest management, and timely harvesting.

68. How can rice yield be improved?

Yield depends on variety, planting time, plant population, nutrient management, water management, weed control and pest/disease management. ICAR specifically identifies balanced fertilizers, suitable varieties, timely planting and integrated nutrient management among important considerations.

([ICAR-CIWA](#))

69. How is maize cultivated?

Maize requires appropriate variety selection, suitable soil preparation, timely sowing, proper spacing, balanced nutrition, adequate moisture and management of weeds and pests.

70. How are pulses grown?

Pulses such as chickpea, lentil and pigeon pea are legumes. They can contribute to diversified cropping systems and, through biological nitrogen fixation, have important roles in nutrient cycling.

71. Why are millets important?

Millets are diverse cereal crops that can provide food and nutritional benefits and are often suited to relatively challenging production environments. Their suitability still depends on local climate, soil and management.

72. How is cotton cultivated?

Cotton production requires suitable varieties, proper sowing, plant spacing, nutrient and water management, weed control, monitoring of insect pests and timely harvesting.

73. How are vegetables cultivated?

Vegetable production involves crop-specific variety selection, nursery management where required, soil preparation, spacing, irrigation, nutrient management, pest/disease monitoring, harvesting and post-harvest handling.

74. How can tomato production be improved?

Important considerations include healthy seedlings, suitable variety, appropriate spacing, balanced nutrition, irrigation management, staking where appropriate, pest and disease monitoring and careful harvesting.

75. How can farmers improve onion production?

Good seed/planting material, suitable planting time, proper spacing, nutrient and irrigation management, weed control, disease and pest monitoring, and proper curing after harvest are important.

I. Horticulture

76. What is horticulture?

Horticulture is the branch of agriculture concerned with fruits, vegetables, flowers, ornamentals, spices and related crops.

77. What is protected cultivation?

Protected cultivation involves growing crops in structures such as greenhouses, polyhouses or net houses where environmental conditions can be partially controlled.

FAO highlights protected cultivation as a method that can help manage environmental conditions, pests and resource use when appropriately designed. ([FAOHome](#))

78. What is greenhouse farming?

Greenhouse farming grows crops inside a protected structure that can modify environmental conditions such as temperature, humidity, light and pest exposure.

79. What is hydroponics?

Hydroponics is a system of growing plants without conventional soil, with nutrients supplied through a water-based growing solution. It requires careful management of nutrients, water quality, pH and environmental conditions.

80. What is vertical farming?

Vertical farming produces crops in vertically arranged growing systems, often indoors or in protected environments. It can make efficient use of space but may require substantial infrastructure and energy.

J. Livestock & Integrated Farming

81. What is integrated farming?

Integrated farming combines multiple farm enterprises such as crops, livestock, fishery, poultry or horticulture so that outputs from one enterprise can potentially support another.

82. What is dairy farming?

Dairy farming involves keeping milk-producing animals and managing feeding, housing, breeding, health, milking and marketing.

83. What should livestock be fed?

Feed should provide adequate energy, protein, minerals, vitamins, fibre and water according to the animal's species, age, production stage and health condition. Local veterinary or livestock-extension advice is important for specific rations.

84. Why is clean water important for livestock?

Water is essential for digestion, temperature regulation, metabolism and milk production. Animals should have regular access to clean, suitable drinking water.

85. What is poultry farming?

Poultry farming involves raising birds such as chickens for eggs or meat. Important management areas include housing, nutrition, biosecurity, disease prevention and appropriate environmental conditions.

K. Farm Economics & Marketing

86. How can a farmer calculate production cost?

Production cost can include:

- Seed
- Fertilizer
- Crop protection
- Labour
- Irrigation
- Machinery
- Fuel/electricity
- Land-related expenses
- Interest or financing costs
- Post-harvest expenses
- Transportation

Keeping accurate records helps farmers understand profitability.

87. How can farmers increase farm income?

Possible strategies include improving productivity, reducing avoidable input costs, diversification, value addition, better post-harvest management, direct or improved market access and selecting enterprises suited to local demand.

88. What is value addition in agriculture?

Value addition means processing or improving a farm product so that it can be sold in a higher-value form. Examples include turning milk into yoghurt, fruit into processed products or grains into packaged products.

89. Why do farmers face post-harvest losses?

Losses can occur because of improper harvesting, drying, storage, transportation, pests, microorganisms, temperature and inadequate packaging.

90. Why should grains be dried before storage?

Drying reduces moisture to an appropriate storage level, which can reduce the risk of mould, microbial growth and deterioration. The safe moisture level varies by commodity and storage conditions.

L. Modern & Smart Agriculture

91. What is precision agriculture?

Precision agriculture uses data, sensors, GPS, remote sensing and other technologies to manage agricultural inputs and operations more precisely according to field conditions.

92. How can AI help farmers?

AI can potentially assist with:

- Crop and variety information
- Pest and disease identification
- Weather interpretation
- Irrigation decisions
- Farm planning
- Market information
- Yield analysis
- Agricultural education

However, AI-generated recommendations should be checked against reliable agricultural and local expert guidance, particularly for high-stakes crop-protection decisions.

93. What is smart farming?

Smart farming combines agricultural knowledge with technologies such as sensors, IoT, AI, data analytics and precision agriculture to improve decision-making and resource efficiency. FAO describes smart farming as combining digital technologies with sustainable farming and resource management. ([FAOHome](#))

94. How can drones help agriculture?

Agricultural drones can be used for field imaging, crop monitoring, mapping and certain targeted applications where legally permitted and technically appropriate.

95. How can satellites help farmers?

Satellite imagery can provide information about vegetation, crop development, field variability, water stress and land conditions over large areas.

96. What is IoT in agriculture?

The Internet of Things connects sensors and devices so they can collect and exchange information. Agricultural applications can include soil-moisture monitoring, weather stations, irrigation systems and livestock monitoring.

M. Sustainable & Future Farming

97. What is organic farming?

Organic farming is a production approach that emphasizes ecological processes and restricts or excludes certain synthetic inputs according to the relevant organic standards. Certification requirements differ by country and standard.

98. What is natural farming?

Natural farming refers to approaches that emphasize ecological processes and reduced reliance on externally purchased agricultural inputs. The exact practices differ among systems and regions.

99. How can agriculture adapt to climate change?

Possible approaches include:

- Climate-resilient varieties
- Adjusting planting dates
- Efficient irrigation
- Water harvesting
- Soil-moisture conservation
- Crop diversification
- Improved soil fertility
- Reduced soil disturbance where appropriate
- Better pest monitoring

FAO identifies changes in varieties, planting dates, irrigation management, soil fertility and pest monitoring among adaptation approaches. ([FAOHome](#))

100. What is the future of agriculture?

The future of agriculture is likely to combine traditional agricultural knowledge with improved genetics, better soil and water management, precision agriculture, AI, sensors, automation, protected cultivation, improved supply chains and climate-resilient practices.

The central goal is not simply "more technology"; it is **better decisions, efficient resource use, resilient farms and sustainable food production**. FAO's current smart-farming work similarly emphasizes digital technologies, efficient use of water and inputs, productivity and ecosystem health. ([FAOHome](#))